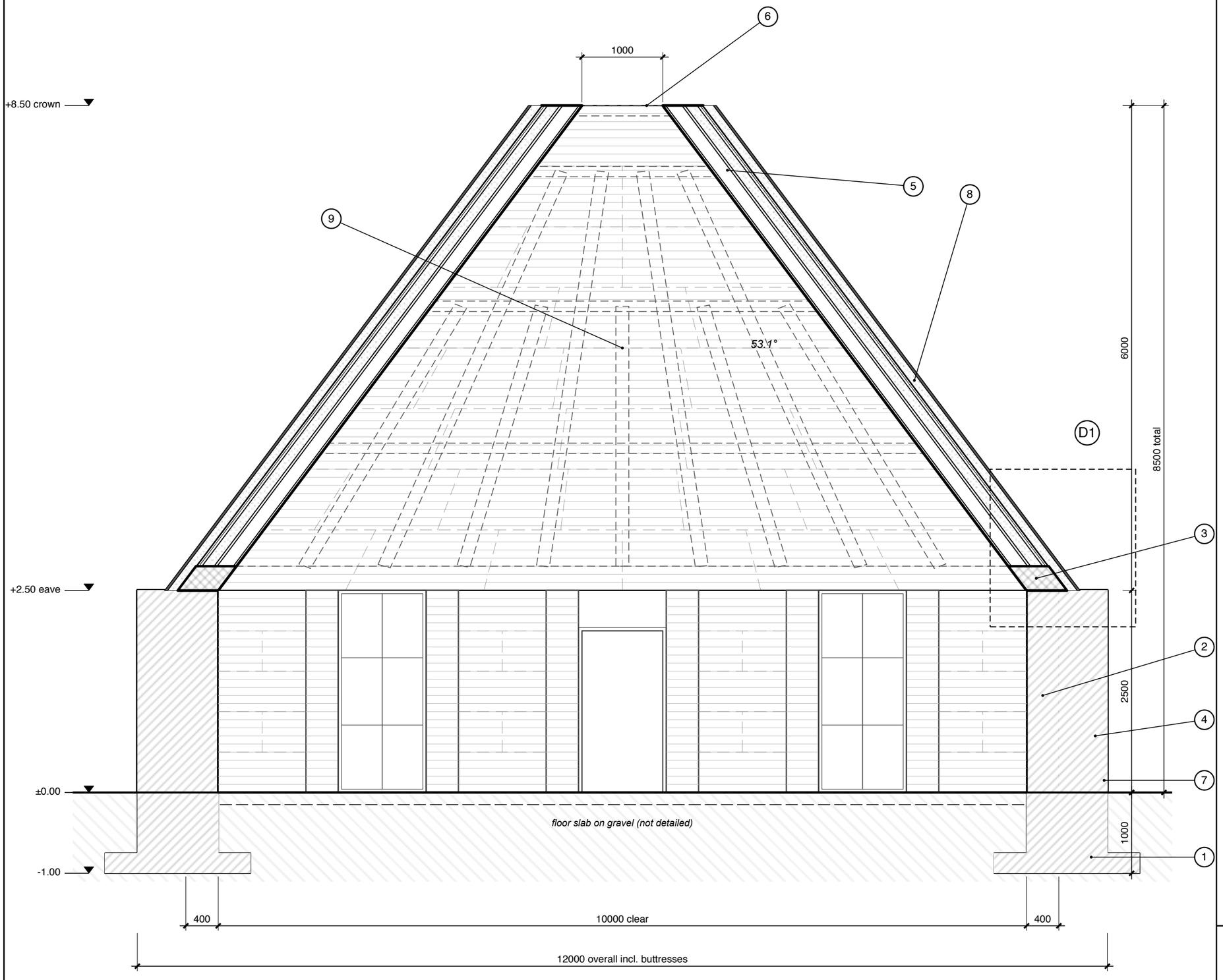
# Total section A-A · grid A

through the east and west buttresses, looking north




1. Footing 1800 x 250, underside at -1000 (frost depth and bearing are guesses).
2. Wall 400. The cut passes through a buttress, so the wall is solid with the buttress here (no insulation). Between buttresses it is printed: an inner bead 30, a lattice zone 310 (wavy bead + insulation), 2 outer beads 30.
3. Solid base of the roof: the first 300 (vertical) above the eave is completely solid concrete, 400 wide, no cavity or lattice zone; there is no ring beam.
4. Buttress 400 wide, 1000 from the inner face (600 beyond the wall), one about every 1.5 m (8 per side); it rises to +2.50, flush with the roof base.
5. Roof shell 400 thick square to the slope, the same from eave to crown (about 500 horizontally), every printed layer one 30 bead: inner bead 30, rib zone 160 (cavity crossed by the cast ribs), middle bead 30, lattice zone 120 (wavy bead + insulation, dotted), 2 outer beads 30.
6. Crown opening 1000 clear; roof rises 6000 over 4500, slope 53.1° (36.9° from the vertical).
7. Everything is 3D printed in 10 mm layers; crossed hatch = solid cast concrete (roof base, ribs). Printed layer joints and the cast fill of the buttress and footing are shown in the roof-grid sheets and in detail D1.
8. Outside the printed shell: insulation 100 (dotted), then the outer layer (cladding 30, dark).
9. Cast ribs of the roof grid, seen on the back face as dashed outlines (roof grid A, ruled-surface columns). The cut lies between ribs, where the rib zone is an empty cavity; rib sections are on the roof-grid sheets.

## Notes

Roof build-up as in the Karamba model (scripts/karamba_model.py): bead 30, 2 outer beads, lattice 120, rib zone 160; inner and middle bead are the structural skins, the ribs the webs between them.

The cut runs through the 4th buttress from the south (y = -743), as on the floor plan. Beyond it the north wall is seen as on the plan: buttress every 1.5 m (strips outside the wall), every other bay printed wall, the bays in between full-height curtain wall (door in the middle one).

Faint grid on the back roof face and the north wall = printed pieces in running bond, about 1500 wide (assumed sizes). Dashed outlines on the back roof face = cast ribs.

0 1 2 3 4 m

**Pyramid roof · 3D-printed concrete · GSAPP**

Scale 1:50 at A3 · dimensions in mm

2026-10-14 · hand estimates, not engineered

**04**